

UCT41042 – Practical for Introduction to Smart System
Department of Information & Communication Technology
Faculty of Technology
South Eastern University of Sri Lanka
Lab Sheet No: 10

Date: 18.06.2026

Title: AI-Based Image Classification using Teachable Machine and Web-based Implementation.

Aims:

To design and implement an AI-based image classification system using Google Teachable Machine, enabling real-time image recognition through a web-based interface. This experiment demonstrates fundamental concepts of machine learning, model training, deployment, and client-side inference using TensorFlow.js.

Task 01: Teachable Machine Model Development

1. Project Setup
 - a. Open the Teachable Machine website
 - b. Select Image Project → Standard Image Model.
 - c. Name the project: TeachableMachine_ImageClassifier
2. Data Collection
 - a. Create at least two classes (e.g., Class A and Class B)
 - b. Collect image samples for each class using:
 - i. Webcam capture OR
 - ii. Uploading images
 - c. Ensure each class contains at least 20–30 images for better accuracy
3. Model Training
 - a. Click the Train Model button
 - b. Wait until the training process is completed
 - c. Observe the training progress and accuracy

4. Model Testing

- a. Test the model using live webcam input
- b. Verify that the model correctly classifies objects
- c. Improve the model by adding more training data if necessary

5. Model Export

- a. Click Export Model
- b. Select TensorFlow.js format
- c. Copy the provided model URL

Task 02: Web-Based Implementation

1. Project Setup in VS Code

- a. Create a new project folder
- b. Name it: TeachableMachine_WebApp

2. Interface Development

- a. Create an HTML file
- b. Add:
 - i. Webcam input
 - ii. Prediction display area

3. Model Integration

- a. Import TensorFlow.js library
- b. Load the Teachable Machine model using the exported URL
- c. Implement prediction logic

4. Real-Time Prediction

- a. Capture webcam frames
- b. Run predictions continuously
- c. Display:
 - Predicted class
 - Confidence score

Task 03: Testing and Validation

1. Model Validation

- a. Test with different real-world objects
- b. Record prediction accuracy and errors

2. System Testing

- a. Verify real-time predictions using webcam
- b. Ensure proper model loading and execution

3. Result Observation

- a. Ensure stable and continuous predictions
- b. Analyze system performance and limitations